The Carnival Caper

Whodunnit? · Circles · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Circles	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Circumference, area, arc length, sector area, inscribed angles, tangents and chords.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Big Lou took it, in the Hall of Mirrors, using this method: Hidden in balloons.

The reveal (read aloud): During the finale, **Big Lou** slipped into the **Hall of Mirrors**, hid the Golden Ticket Trophy inside a bunch of **balloons**, and floated it out the door. Every alibi and method you cleared led to him. Right on the circle!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Ringo · Place: Ferris Wheel
2	Method: Stuffed in a prize toy
3	Suspect: Ms. Taffy · Place: Cotton Candy Stand
4	Method: Carried in a candy cone
5	Suspect: Mr. Dunk · Place: Dunk Tank
6	Method: Hauled on a ride cart
7	Suspect: Penny · Place: Ticket Booth
8	Method: Hidden in the popcorn machine
9	Suspect: Clara · Place: Bumper Cars
10	Method: Dropped in a bumper car

Answer Key

The Carnival Caper · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Find the circumference of a circle with radius 5. Leave your answer in terms of π.	**B.** 10π	$C = 2\pi r = 2\pi(5)$ 10π	Ringo was running the Ferris Wheel. His ride log has every loading time.
2	Find the area of a circle with radius 6. Leave your answer in terms of π.	**A.** 36π	$A = \pi r^2 = \pi(6^2)$ 36π	The prize toys were all checked at the exit. None held the trophy.
3	Find the area of a circle with diameter 14. Leave your answer in terms of π.	**C.** 49π	$r = 7$ $A = \pi(7^2)$ 49π	Ms. Taffy was spinning cotton candy at her stand with a long line of kids.
4	A central angle measures 80°. Find the measure of an inscribed angle that intercepts the same arc.	**D.** 40°	inscribed angle $= \frac{1}{2} \cdot$ arc $40°$	The candy cones are all empty paper and sugar. Nothing inside.
5	An inscribed angle measures 35°. Find the measure of its intercepted arc.	**A.** 70°	arc = 2 · inscribed angle 70°	Mr. Dunk was getting dunked in the Dunk Tank for charity. Hundreds watched.
6	Point P is 13 units from the center O of a circle with radius 5. A tangent from P touches the circle at T. Find PT.	**B.** 12	the radius OT is perpendicular to the tangent PT $PT^2 + 5^2 = 13^2$ $PT = 12$	The ride carts were wheeled back empty and checked.
7	Find the length of an arc with radius 9 and central angle 40°. Leave your answer in terms of π.	**B.** 2π	arc $= \frac{40}{360} \cdot 2\pi(9)$ 2π	Penny was selling tickets at the Ticket Booth. Her ticket roll shows each sale.
8	Find the area of a sector with radius 6 and central angle 60°. Leave your answer in terms of π.	**C.** 6π	sector $= \frac{60}{360} \cdot \pi(6^2)$ 6π	The popcorn machine was opened. Just popcorn.
9	Two chords intersect inside a circle. One chord is split into pieces of length 3 and x, and the other into pieces of length 4 and 6. Find x.	**D.** 8	$3 \cdot x = 4 \cdot 6$ $3x = 24$ $x = 8$	Clara was painting faces at the Bumper Cars with a line of kids waiting.
10	A circle has circumference 18π. Find its radius.	**D.** 9	$2\pi r = 18\pi$ $r = 9$	The bumper cars were all inspected. Nothing was hidden in any of them.

Wrong-Answer Map

The Carnival Caper · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. 25π	used area formula	Big Lou was waiting in line for the Ferris Wheel with a bunch of balloons. The operator remembers him.
1	C. 5π	forgot the 2	Ringo was tightening bolts near the Hall of Mirrors. Two kids asked him for directions.
1	D. 20π	used diameter as radius in 2πr... doubled	Every balloon bunch was popped and checked. Nothing inside but air.
2	B. 144π	used diameter squared	The balloons by the gate were all squeezed. No bulge, no trophy.
2	C. 12π	used 2πr	Big Lou was buying a cotton candy at the Cotton Candy Stand. Ms. Taffy handed it over.
2	D. 6π	used πr	Ms. Taffy was straightening the mirrors in the Hall of Mirrors. A group of kids counted her reflections.
3	A. 196π	used diameter as radius	Big Lou was trying his luck at the Dunk Tank. He got soaked and the crowd cheered.
3	B. 14π	circumference with diameter	Mr. Dunk was fixing a latch in the Hall of Mirrors. His toolbox log shows the time.
3	D. 7π	used r times π	The balloon cart was unloaded and every balloon examined. All ordinary.
4	A. 160°	doubled instead of halved	Penny was counting change in the Hall of Mirrors on her break. A guard saw her go in.
4	B. 80°	inscribed equals central	Big Lou was asking Penny for a ticket at the Ticket Booth. Her roll shows his purchase.
4	C. 20°	halved twice	A balloon bouquet was slit open at the exit. Only string and helium.
5	B. 17.5°	halved instead of doubled	The helium tank and every balloon were inspected. Nothing was tucked in any of them.
5	C. 140°	doubled twice	Clara was painting a star on a kid's cheek in the Hall of Mirrors. The kid still has it.
5	D. 35°	arc equals inscribed angle	Big Lou was bumping around the Bumper Cars with his nephew. The ride photo proves it.
6	A. 18	added lengths	Not a single balloon felt heavy. The trophy could not be hidden in one.
6	C. √194	added squares	Big Lou was riding the Ferris Wheel at the top when the lights flickered. A photo caught him waving.
6	D. 8	subtracted lengths	Clara was giving a butterfly to a kid in the Hall of Mirrors. Her brush is still damp.
7	A. 18π	forgot angle fraction	Ringo was announcing the next ride from the Hall of Mirrors. The loudspeaker clip is timed.
7	C. 9π	found sector area	The balloon bunches were weighed one by one. All were light as feathers.
7	D. π	used diameter as radius	Big Lou was handing out balloons next to the Dunk Tank. A line of kids saw him.
8	A. 2π	arc length instead of area	A balloon was popped on purpose at the gate. Just a puff of air came out.

8	B. 36π	full circle area	Big Lou was sharing a pink cloud of sugar at the Cotton Candy Stand. His sticky hands told the story.
8	D. 12π	full circumference	Mr. Dunk was practicing his dunking pitch near the Hall of Mirrors. A passerby recorded it.
9	A. 4.5	cross-multiplied wrong pieces	Ms. Taffy was refilling sugar bags and crossed through the Hall of Mirrors. The security tape shows it.
9	B. 7	added instead of multiplied	The balloon man's bundle was checked by the fair manager. No trophy inside.
9	C. 24	forgot to divide by 3	Big Lou was waiting at the Ticket Booth for a prize ticket. Penny stamped his hand.
10	A. 4.5	halved twice	Every balloon in the park was counted and squeezed. Only balloons, no trophy.
10	B. 18	gave diameter	Big Lou was selling balloons beside the Bumper Cars all afternoon. A dozen kids bought one.
10	C. 36	doubled again	Penny was chasing a runaway ticket stub through the Hall of Mirrors. A clerk caught it.